Baseline ATL V1 Validation Update

12 October 2026 | Prepared by Haroon | Correction and evidence update

Release decision: NOT PRODUCTION READY. The two original security fixes remain completed. Real Stripe decline and failure-event delivery evidence is now saved. Production validation, operating controls and business decisions remain open. This update does not provide launch sign-off.

Consistent replacement for Sections 2.3 and 3

At submission, the correct status for both disputed rows was **Built but not verified**. The claims in Section 2.3 were unsupported. New dated evidence below now permits the same scoped status in both sections:

Scenario	Revised status and evidence scope
Failed payment in Stripe test mode	**Built and verified** for insufficient-funds and fraudulent-card declines on staging, 12 October. Both remained unpaid; capacity was held until cancellation and restored afterwards.
Real Stripe webhook delivery to staging	**Built and verified** for real failure-event delivery and HTTP acknowledgement, 12 October. Stripe events reached zero pending webhooks; saved staging request logs show HTTP 200. This does not newly verify a successful-payment webhook registration.

Mock decline checks and automated signature/replay checks retain their verified status within their local test scope. They are separate from the real-provider evidence above.

Recorded results and completed work

Staging, 09:48-09:52 PKT: Two real Stripe decline cases passed on an isolated two-place QA league. A repeat run verified concurrent same-player retries reused the reservation, intent and expiry; authenticated refresh found the original hold. Capacity changed 2 to 1 while held, stayed 1 after decline, and returned to 2 after cancellation. No real money was charged.

Webhook evidence: Two Stripe-generated payment_failed events and four staging webhook requests are saved. Event pending_webhooks fell from 2 to 0. Runtime logs correlate by timestamp; they do not contain event IDs, and no successful-payment registration claim is made from these failure events.

Security: Organiser-only creation and exact-origin CORS were pushed in 19e908b. A further confirmation flaw found during this review is fixed in 071e114: client data cannot fabricate a paid registration; the API checks ownership and provider state, and the UI waits for server-confirmed registration. Both commits are on backend/staging-load-test-v1.

Local checks: 33 backend tests, 7 backend unit tests and 27 frontend tests passed. TypeScript checks and the frontend production build passed. Backend integration tests use an in-memory MongoDB replica set. The new confirmation fix has not been verified on the current manual staging build.

Build identity and cleanup: Staging deployment dpl_jqV5yDjnLt4BR2AuPc8WejUU5RWi is labelled f7c3759 and differs from the Git release source. No replacement deployment was made. The QA league has zero active reservations, restored capacity and is closed; seven new QA accounts are disabled. Financial/audit records are retained. Shared demo data was not bulk-deleted.

Decisions and remaining release gates

The current choices are Atlas Free and Vercel Hobby. Ali should approve the changes below before engineering claims production readiness.

Hosting and database approach

Vercel: Hobby allows cron at most daily, with execution within the scheduled hour, and is restricted to personal non-commercial use. The repository daily jobs cannot guarantee release within minutes after a 15-minute hold. Approve Pro or eligible commercial hosting first, then deploy expiry every 2 minutes and reconciliation every 5 minutes, save actual invocations and test missed-job alerts. Do not deploy unsupported frequent cron to Hobby. Pro currently has a $20/month platform fee including one deploying seat and $20 usage credit; extra seats, add-ons and usage can add cost. Approve a spending limit. [1-3]

Atlas: Free has no managed backups; MongoDB documents manual mongodump/mongorestore as an alternative. The submitted 500-user tests had 8-19 second p95 times and errors in two of three runs. Ali should approve peak concurrency, latency/error targets, acceptable data loss and recovery time, then select a paid candidate with the required backup capability and repeat representative deployed tests. For continued Free-tier testing, arrange encrypted off-site dumps and an isolated restore drill. Neither a tier upgrade nor a backup file alone proves readiness. [4]

Remaining gap	Required action and closure evidence
Full Gate A and release build **Built but not fully verified**	Preserve newer source changes, reconcile the manual deployment with Git, deploy the approved build and rerun expiry/race and payment-at-expiry cases on staging and the chosen production tier. Retry/refresh passes cover only the stated scenarios.
Refund workflow **Partially built**	Ali approves full/partial eligibility, organiser authority and spot-release rules. Engineering builds a retry-safe organiser refund endpoint and verifies provider outcomes and capacity. A refund must not implicitly free a spot without the approved policy.
Live merchant and payment mode **Decision blocked**	Ali confirms provider, merchant entity, settlement/currency and agreement; then authorises a controlled live test. Record effective provider and test/live mode on every deployed environment without exposing keys.
Recovery and monitoring **Not complete**	Approve backup retention and recovery targets; restore into an isolated database and verify counts/indexes/capacity. Assign an alert owner, extend log retention and save received job/payment/database/availability alerts.
Email and support **Not built in submitted record**	Choose provider and authenticated sender, implement confirmation/receipt delivery with retries and duplicate suppression, and save delivery evidence. Ali supplies a monitored support address and owner.
Runbook and secrets **Procedure prepared**	The companion runbook covers redeploy, rollback, evidence and secret ownership. Execute and record the drill; assign secret owners and rotation/revocation procedures. Documentation is not a completed recovery test.
Data cleanup and final sign-off **Open**	Verify a backup; approve exact synthetic-record cleanup, preserve financial/audit history and reconcile capacity. Complete the production smoke journey with a reviewer and obtain release approval.

Official sources checked 10-12 October 2026: [1] Vercel cron limits; [2] Hobby use restrictions; [3] Pro pricing; [4] Atlas Free backup limits. No plan upgrade, merchant decision or broad data wipe was performed.

Evidence 1 Insufficient funds decline

12 October 2026. Real Stripe test API result plus authenticated staging state. The held place remains reserved after decline; cancellation restores capacity. Concurrent retry and refresh assertions also pass.

Real Stripe decline validation

Actual staging requests and Stripe test API results. No real-money transaction. Saved 12 October 2026.

Environment: https://atlanta-tennis-staging.vercel.app
Run: 2026-10-12T04:51:18.668Z to 2026-10-12T04:51:54.114Z
League: qa-gap-validation-20261012
Capacity: 2 before / 2 after cleanup
Overall pass: true

insufficient-funds


```
{
  "reservationId": "6acc674ac24f1b1cf9bfd166",
  "paymentIntentId": "pi_3UPbCp3LwR6lzVYP0ykHrxzu",
  "declineCode": "insufficient_funds",
  "stripeStatus": "requires_payment_method",
  "livemode": false,
  "applicationStatus": "payment_pending",
  "paymentStatus": "pending",
  "spotsHeld": 1,
  "spotsAfterDecline": 1,
  "spotsAfterCleanup": 2,
  "retryAssertions": {
    "sameReservation": true,
    "sameIntent": true,
    "holdNotExtended": true
  },
  "refreshFoundOriginal": true,
  "assertions": {
    "declineMatches": true,
    "testMode": true,
    "notRegistered": true,
    "paymentNotPaid": true,
    "slotRetained": true,
    "providerEventExists": true
  }
}
```


Screenshot captured from a local read-only viewer of the actual saved records. Raw evidence is retained in docs/evidence/2026-10-12. This is not a Stripe Dashboard screenshot.

Evidence 2 Fraudulent card decline

12 October 2026. Real fraudulent-card decline; livemode is false. The application remains payment_pending, and the same reservation and intent survive retries without extending the hold.

Real Stripe decline validation

Actual staging requests and Stripe test API results. No real-money transaction. Saved 12 October 2026.

Environment: https://atlanta-tennis-staging.vercel.app
Run: 2026-10-12T04:51:18.668Z to 2026-10-12T04:51:54.114Z
League: qa-gap-validation-20261012
Capacity: 2 before / 2 after cleanup
Overall pass: true

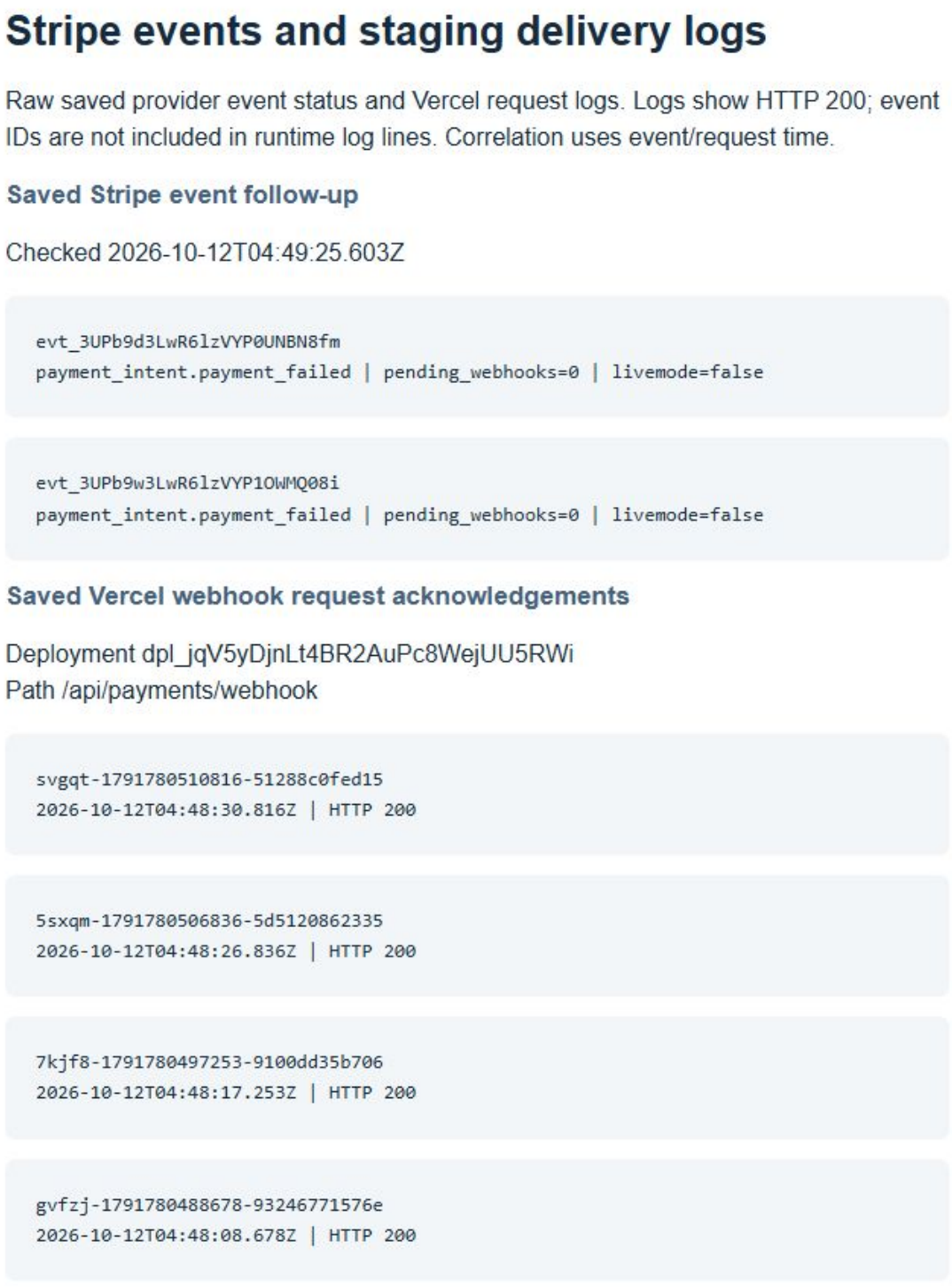
fraudulent


```
{
  "reservationId": "6acc675bc24f1b1cf9bfd169",
  "paymentIntentId": "pi_3UPbD53LwR6lzVYP0itmeOqy",
  "declineCode": "fraudulent",
  "stripeStatus": "requires_payment_method",
  "livemode": false,
  "applicationStatus": "payment_pending",
  "paymentStatus": "pending",
  "spotsHeld": 1,
  "spotsAfterDecline": 1,
  "spotsAfterCleanup": 2,
  "retryAssertions": {
    "sameReservation": true,
    "sameIntent": true,
    "holdNotExtended": true
  },
  "refreshFoundOriginal": true,
  "assertions": {
    "declineMatches": true,
    "testMode": true,
    "notRegistered": true,
    "paymentNotPaid": true,
    "slotRetained": true,
    "providerEventExists": true
  }
}
```


Screenshot captured from a local read-only viewer of the actual saved records. Raw evidence is retained in docs/evidence/2026-10-12. This is not a Stripe Dashboard screenshot.

Evidence 3 Real webhook delivery

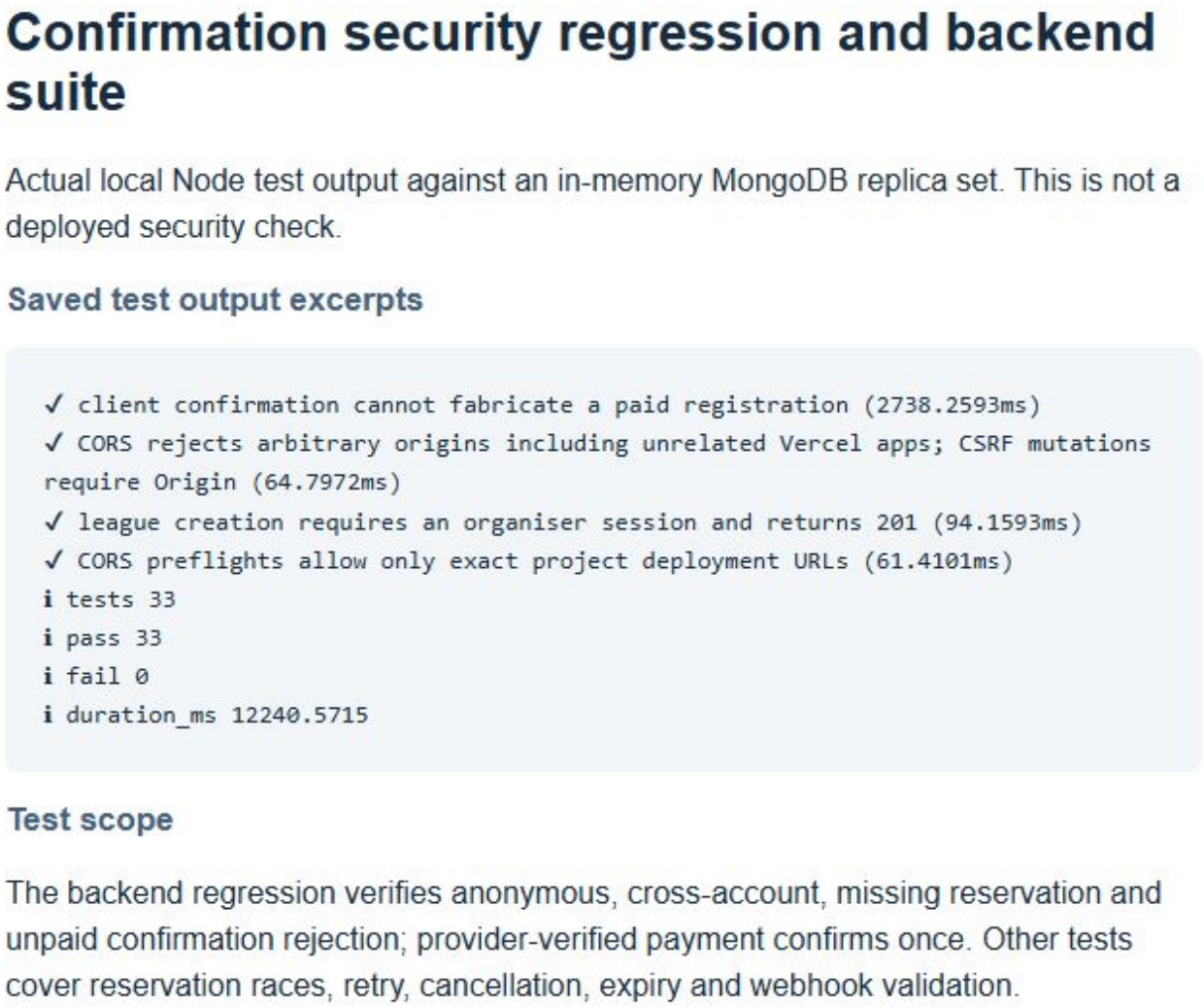
12 October 2026, initial run at 09:48 PKT. Stripe failure events have no pending deliveries at follow-up; staging records show four HTTP 200 webhook acknowledgements. Correlation is by time, not event IDs in log lines.



Screenshot captured from a local read-only viewer of the actual saved records. Raw evidence is retained in docs/evidence/2026-10-12. This is not a Stripe Dashboard screenshot.

Evidence 4 Security regression tests

12 October 2026. Local integration suite passes all 33 tests, including rejection of fabricated client confirmation and the original organiser/CORS checks. This screenshot does not prove deployment of the new confirmation fix.



Screenshot captured from a local read-only viewer of the actual saved records. Raw evidence is retained in docs/evidence/2026-10-12. This is not a Stripe Dashboard screenshot.

Raw companion records include stripe-declines.json, stripe-declines-initial.json, webhook-event-followup.json, webhook-delivery-logs.json, backend-tests.txt, backend-unit-tests.txt, frontend-tests.txt, frontend-typecheck.txt, frontend-build.txt and qa-cleanup.json.